



PROJECT-BASED LEARNING

PHASE-I EVALUATION

Smart Emergency Response System

Team ID: DSCPP-III-2026-T232

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026-27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T232
Semester: 3rd
Section: DS1, ML1, B, DS2
Domain: DSA / C++

Team Members

1. Sambhawi Pandey – 2510380216
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Mentor

Mr.Rohan Verma

Interactions completed: 3

Problem Statement & Motivation

Problem Statement

Emergencies such as road accidents, fires and medical incidents may occur simultaneously while emergency resources are limited. Manual prioritization and resource allocation can delay response.

Motivation

- ▶ Faster emergency prioritization
- ▶ Efficient resource allocation
- ▶ Shortest route selection
- ▶ Handling multiple emergencies

Target Users / Stakeholders

- ▶ Emergency operators / dispatchers
- ▶ Ambulance and response teams
- ▶ Hospitals, police and fire services
- ▶ Emergency reporting users

Objectives

Project Objectives

1. Develop an emergency reporting and management system.
2. Categorize and prioritize emergencies using severity and Priority Queue.
3. Manage emergency resources such as ambulances efficiently.
4. Model roads and calculate shortest routes using Graphs and Dijkstra's Algorithm.
5. Apply DSA and OOP in C++ and test the system with different emergency scenarios.

Key Objective: Emergency response system that prioritizes incidents, manages resources, and finds optimal routes

Proposed Solution

Proposed Approach

1. Emergency Data Collection
2. Emergency Validation
3. Severity-Based Prioritization
4. Resource Allocation
5. Shortest Route Calculation
6. Dispatch & Status Update

Key Features

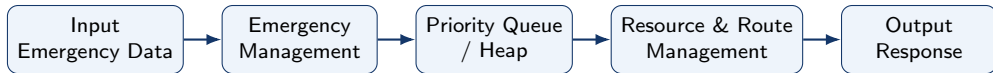
- ▶ Emergency Management
- ▶ Priority Queue / Heap
- ▶ Ambulance Management
- ▶ Graph + Dijkstra
- ▶ Status Tracking
- ▶ Searching & Sorting

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	Arrays, Linked Lists, Queues, Priority Queue, Heaps, Graphs, Searching & Sorting	Emergency & Resource Management
OOPS in C++	Classes, Objects, Encapsulation, Abstraction, Constructors and OOP Concepts	Emergency & Ambulance Modules

Core Integration: Data Structures provide the core algorithms and data handling, while C++ is used to implement the complete system.

System Architecture / Workflow



Major Components

- ▶ Emergency Management
- ▶ Priority Queue / Heap
- ▶ Resource Management
- ▶ Graph + Dijkstra's Algorithm

Data / Control Flow

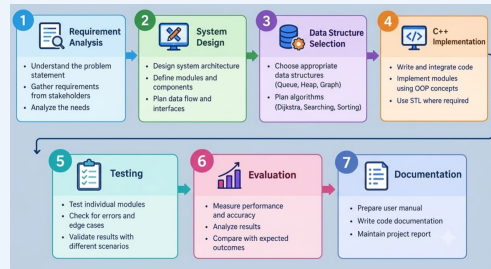
- ▶ Emergency data is registered and categorized by severity.
- ▶ Priority Queue prioritizes critical emergencies.
- ▶ Available resources are selected and shortest routes are calculated.
- ▶ System displays assigned resource, route and response status.

Technology Stack & Methodology

Technology Stack

- ▶ Programming Language: **C++**
- ▶ Paradigm: **Object-Oriented Programming**
- ▶ Data Structures: **Queue, Heap, Graph**
- ▶ Algorithms: **Dijkstra, Searching, Sorting**
- ▶ Development Tool: **Visual Studio Code**
- ▶ Version Control: **Git / GitHub**

Development Methodology



Initial Progress & Team Contribution

Progress Achieved

- ▶ Background / literature study completed
- ▶ Requirements identified
- ▶ System architecture prepared
- ▶ Initial modules planned
- ▶ Data requirements identified

Individual Contribution

Member	Role	Contribution
Sambhawi Pandey	Team Lead	Planning, coordination and system design
Atharv Rawat	Developer	Core C++ modules and algorithms
Aayush Lalotra	DS / Testing	Data structures and test cases
Shivam Garg	Developer/ Docs	Implementation support and documentation

Roadmap & Expected Outcomes

Roadmap

1. Phase-I: Proposal & Design
2. Phase-II: Development
3. Phase-III: Final Implementation

Expected Outcomes

- ▶ Emergency prioritization
- ▶ Resource allocation
- ▶ Shortest route calculation
- ▶ Emergency status tracking
- ▶ Testable system results

References

1. "Ambulance Routing for Disaster Response with Patient Groups," *Computers & Operations Research*, vol. 56, pp. 120–133, 2015.
2. "Integrating GIS, GPS and GSM Technologies for the Effective Management of Ambulances," *International Journal of Medical Informatics*.
3. "A Systematic Review of Route Optimisation and Pre-emption Methods for Emergency Vehicles," *Transport Reviews*, vol. 40, no. 1, pp. 35–53, 2020.

Thank You

Questions & Suggestions